

Program : Diploma in Polymer Technology	
Course Code : 5097	Course Title: Compound Design Workshop
Semester : 5	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To design and develop rubber compounds with different properties.
- To analyze the properties of different rubber products.
- To get hands-on experience in physical testing of various rubber products.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics 1 & II	1 & 2
Basic Knowledge in chemistry		Applied Chemistry	I
Rubber Compounding		Technology of Elastomers	3
Properties of Synthetic rubbers		Synthetic Rubbers	3

Course Outcomes:

On completion of the course ,the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Prepare gum rubber compounds and study the hardness.	6	Applying
CO2	Practice designing of NR-HAF vulcanizates compounds with specific properties	17	Applying
CO3	Practice designing of SBR-Silica vulcanizates compounds with specific properties	18	Applying
CO4	Develop rubber compound for pre-determined specific gravity.	2	Applying
	Lab exam	2	

CO- PO mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3		3			
CO2	3	3		3			
CO3	3	3		3			
CO4	3	3		3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Prepare gum rubber compounds and study the hardness.		
M1.01	Prepare gum compounds with NR and measure the hardness	3	Applying
M1.02	Prepare gum compounds with Synthetic Rubber and measure the hardness.	3	Applying
CO2	Practice designing of NR-HAF vulcanizates compounds with specific properties		
M2.01	Design NR compounds of specific hardness (40, 50, 60, 70 Shore A) using HAF as filler.	3	Applying
M2.02	Preparation of NR- HAF vulcanisates and study their cure characteristics.	6	Applying
M2.03	Preparation of test samples for hardness and tensile properties of NR- HAF vulcanizates.	3	Applying
M2.04	Analyse the effect of varying filler loading on hardness of NR-HAF vulcanisates	3	Applying
M2.05	Analyses the effect of varying filler loading on tensile properties (TS, M300 EB) of NR- HAF vulcanisates	2	Applying
	Lab Exam I	1	
CO3	Practice designing of SBR-Silica vulcanizates compounds with specific properties		
M3.01	Design SBR rubber compounds of specific hardness (40, 50, 60, 70 Shore A) using Silica as filler.	3	Applying
M3.02	Preparation of SBR- Silica vulcanizates and study their cure characteristics.	6	Applying
M3.03	Preparation of test samples for hardness and tensile properties of SBR- Silica vulcanizates.	3	Applying

M3.04	Analyse the effect of varying filler loading on hardness of SBR-Silica vulcanisates	3	Applying
M3.05	Analyses the effect of varying filler loading on tensile properties (TS, M300 EB) of SBR-Silica vulcanisates	3	Applying
CO4	Develop rubber compound for pre-determined specific gravity.		
M4.01	Design NR compounds for pre-determined specific gravity using Carbon black filler and test the specimen	2	Applying
	Lab exam II	1	

Text / Reference:

T/R	Book Title/Author
T1	Rubber Technology and Manufacture - C.M.Blow, (publisher-Butterworth Hainemann Ltd,1982 Edition
R2	Rubber Technology - Maurice Morton (Publisher - Springer 1999 Edition)
R3	Rubber Technology Handbook - Werner Hofman (Publisher - Oxford University Press,1989 edition
R4	Vanderbilt Handbook - R.T. Vanderbilt,1990 - R.T Vanderbuilt,1990 edition

Online Resources:

Sl. No	Website Link
1	https://www.ncbi.nlm.nih.gov/pubmed
2	https://www.tandfonline.com
3	- https://www. Researchgate.net & publication
4	https://www.onlinelibrary.wily.com
5	https://www.sciencedirect.com